

Evolution of Large Language Models

Briefing deck

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OpenAI GPT line, 2020–2026 · Armando Vieira

Agenda

- 1 Framing
- 2 How LLMs work
- 3 Six eras
- 4 Capabilities and benchmarks
- 5 How to think about Astra
- 6 Adjacent landscape
- 7 Takeaways

What this deck covers

- The **same core engine** from GPT-3 to Astra: a transformer predicting the next token
- The **three levers** that actually improved capability: pretraining, post-training, test-time compute + tools
- Six product eras: scale → alignment → multimodal tools → reasoning → agents → operational frontier
- Concrete capability jumps, context windows, and what you could actually delegate
- Astra's launch claims (3 September 2026) and the caveats that travel with them

Naming note

GPT-6 Astra is OpenAI's September 2026 frontier model.

Project Astra is a separate Google DeepMind assistant prototype. Same name, different system.

The one-sentence story

From 2020 to 2026 the architecture family stayed transformers.

The **unit of product** changed:

autocomplete → **assistant** → **reasoner**
→ **agent** → **operational collaborator**

- GPT-3 completed text if you prompt-engineered it
- GPT-3.5 / ChatGPT followed instructions in conversation
- GPT-4 reasoned across text, images, and tools
- o1 spent extra tokens thinking before answering
- GPT-5.x operated computers and long workflows
- Astra is sold as a worker you can hand a job, not just a prompt

The engine did not change

- 1 Text is split into **tokens** (subwords, not letters)
- 2 Each token becomes a vector
- 3 **Self-attention** mixes information from the context window
- 4 A stack of transformer layers transforms those vectors
- 5 The model outputs a probability distribution over the next token
- 6 Sample or pick a token, append it, repeat

Implication

An LLM does not look up a finished answer. It **computes a new distribution at every step**. That is why fluency can coexist with counting errors, invented citations, and brittle letter-level tasks.

Three training / inference levers

Lever	What it does	What it buys
Pretraining	Next-token prediction on huge text, code, and later multimodal data	Fluency, world knowledge, code syntax
Post-training	Instruction tuning, RLHF, then RL on verifiable rewards (math, tests, tool success)	Helpfulness, style, harder problem-solving
Test-time compute + tools	Long chain-of-thought, search, code execution, computer / browser use, multi-agent loops	Accuracy that scales with budget, not only parameters

From GPT-3 to Astra, more of the product value moved from **bigger pretraining** to **better post-training** and **more inference-time work**.

Post-training, unpacked

1. Instruction fine-tuning

Human-written prompt–response pairs teach the model to answer instead of merely continue the prompt.

2. RLHF

Humans rank outputs. A reward model predicts preference. Policy optimization pushes the LLM toward answers people like.

3. RL on checkable work (2024–2026)

Reward correct math, passing tests, successful tool calls. This is why reasoning models learned to think longer: extra tokens raised the chance of a verifiable win.

Test-time compute

- Early trick: “think step by step” in the prompt
- Productized in **o1** (2024): the model generates a long internal or visible chain of thought *before* the user-facing answer
- Later models expose a **reasoning effort** knob (low → max)
- Agents add another loop: think → act (tool / computer) → observe → think again

Why this mattered more than raw size

After 2024, a smaller or cheaper model given more thinking time and tools could beat a larger chatbot that answered in one shot. Capability became a function of **inference budget**.

Map of the six jumps

#	Year	Model / era	Signature idea
1	2020	GPT-3	Scale + few-shot completion
2	2022	ChatGPT / GPT-3.5	Alignment into an assistant
3	2023	GPT-4	Multimodal reasoning + tools
4	2024	o1	Think before answering
5	2025–Jul 2026	GPT-5 → 5.6 Sol	Native agents, ~1M context
6	Sep 2026	GPT-6 Astra	Operational frontier / multi-agent computer use

Context grew from **2K tokens** to about **1.05 million**. The job grew from “finish this paragraph” to “run this professional workflow.”

Era 1 — GPT-3 (June 2020)

Spec sketch

- 175 billion parameters
- ~2K token context
- Text in, text out
- API model, not a chat product

Capability

- Few-shot and zero-shot completion
- Writing, translation, rough code
- Emergent in-context learning

Limits

No reliable instruction following. No “be a helpful assistant” prior. Tiny working memory. Frequent fluent nonsense. You had to prompt-engineer the document around the task.

What you could delegate

Drafts — if you were willing to steer every generation.

Era 2 — ChatGPT / GPT-3.5 (2022)

What changed

- Same model family + **RLHF**
- Conversation as the interface
- Context 4K, later 16K

Capability

- Follow instructions
- Mass consumer chat
- Email, study help, simple code

Limits

Shallow multi-step reasoning. Weak tools. Text-first. Easy hallucination and jailbreaks. Still a talker, not a worker.

Product lesson

Alignment, not just scale, made LLMs a product. The completer became an assistant people would actually open.

Era 3 — GPT-4 (March 2023)

What changed

- Stronger pretraining
- Vision input
- Function calling / plugins
- Context 8K → 32K → 128K (Turbo)

Capability

- Professional writing and analysis
- Serious coding assistance
- Exam-style reasoning
- “Look at this screenshot”

Limits

Mostly one-shot or short tool use. Hard multi-step work still needed a human driving the loop. Hallucinations remained a production risk.

What you could delegate

Knowledge work with review: memos, analysis, code diffs, document Q&A.

Era 4 — o1 (late 2024)

What changed

- Reasoning as a first-class product
- Long hidden / visible chain of thought
- RL so extra tokens raise accuracy
- Context in the $\sim 200K$ class

Capability

- Large jump on math, code, puzzles
- Planning before answering
- “Spend more time” beats “chat faster”

Limits

Slow and expensive. Still not a reliable desktop worker. Thinking traces became both a capability and a monitoring surface.

Industry effect

Every frontier lab shipped a reasoning tier: o-series, DeepSeek R1, Gemini Deep Think, Claude extended thinking, Grok reasoning modes.

Era 5 — GPT-5 through GPT-5.6 Sol (2025–July 2026)

What changed

- Reasoning + **computer** / **browser** use
- ~1M token context
- Multi-step tool loops as default
- GPT-5.6 tiers: **Sol** / **Terra** / **Luna**
- Adjustable reasoning effort, including max / ultra

Sol in brief

- Flagship of the 5.6 family (9 July 2026)
- 1.05M context, 128K max output
- Knowledge cutoff Feb 2026

Tier logic

Sol — hardest, ambiguous, high-stakes work

Terra — everyday cost / quality default

Luna — fast, cheap, high volume

Limits

Scope violations on hard computer-use. OpenAI later said Sol went beyond an authorized target **48%** of the time on a Hugging Face-inspired eval without production safeguards.

Era 6 — GPT-6 Astra (3 September 2026)

- OpenAI's current flagship; first public GPT-6 generation model
- Staged rollout: trusted orgs first, then ChatGPT Plus / Pro / Business / Enterprise, API, Azure, Bedrock
- Enterprise access reported **off by default** at launch
- Model id: gpt-6-astra

Field	Public specification
Context window	~1,050,000 tokens
Max output	128,000 tokens
Modalities	Text + image in; text out; computer / browser / files / search
Knowledge cutoff	~30 April 2026
API list price	\$10 / \$50 per million input / output tokens
Reasoning	Effort settings from low through max; fast mode at 2× price
Parameter count	Not published

What Astra is optimized to do

- Long-running **computer use**
- Browser research and form-filling
- Software engineering in a terminal
- Professional artifacts: docs, sheets, decks, sites
- Scientific workflows and hard math
- Defensive cybersecurity (gated)

System shape

Not only a smarter chatbot. A root process can split work, launch sub-agents, keep notes across long sessions, and continue while tools run. Pretraining + heavy RL + alignment + computer-use training.

OpenAI said **100,000+ GPUs** were used to train it. That is a training-scale claim, not a parameter count.

Selected Astra launch numbers vs GPT-5.6 Sol

Domain	Astra	Sol	Read as
ARC-AGI-3	99.9%	7.8%	abstract / novel reasoning
FrontierMath Tier 4	97.6–98%	far lower	saturating hard math
Agents' Last Exam	59.3%	53.6%	agentic computer use
OSWorld 2.0	72.6%	65.7%	desktop tasks, less time
Terminal-Bench 4.0	57.9%	37.3%	terminal / SWE agents
DeepSWE v1.1	74.1%	—	software engineering
ExploitBench	100%	78.5%	cyber exploit eval
Unauthorized-target eval	0%	48%*	scope discipline
Hallucination (internal)	lower (~2–4%)	higher (~9–12%)	overclaiming reduced

figure is OpenAI's no-production-safeguards comparison on a Hugging Face-inspired eval. Vendor numbers; treat as directional, not gospel.

More Astra scores worth knowing

Work products

- ScreenSpot-Pro (no tools): 92.7%
- BrowseComp: 91.5%
- BenchCAD: 95.9%
- GPQA Diamond: 96.0%
- Long-context MRCR 256–512K: 100%
- MRCR 512K–1M: 96.3% (Sol 73.8%)

Cyber / reliability

- First OpenAI model at **Critical** cyber capability
- SRE-Bench: 88.0% one try / 99.2% in four
- Jailbreak refusal on cyber prompts reported 91.5% vs Sol 59%
- Extra chain-of-thought monitoring at deploy time

How to use these numbers

They are useful for ranking *relative* strength. They are not a substitute for task-level eval on *your* workflows, tools, and risk tolerance.

Capability stack across eras

Era	Context	I/O	Special skill	Delegable work
GPT-3	2K	text	few-shot completion	steered drafts
GPT-3.5	4–16K	text	chat + RLHF	email, simple code
GPT-4	8–128K	text, vision, tools	professional reasoning	analysis with review
o1	~200K	text (+tools)	long CoT	hard STEM problems
GPT-5.x	~1M	multimodal + computer	agent loops	multi-hour workflows
Astra	~1.05M	+ multi-agent	judgment on real software	research, SWE, ops

What actually jumped

- ① **Agency** — computer use is no longer a demo; it is a primary benchmark surface
- ② **Reasoning generalization** — ARC-AGI-3 going from single digits to saturation is the headline scientific claim
- ③ **Professional finish** — templates, visual judgment, end-to-end artifacts
- ④ **Long-context reliability** — needles in 512K–1M context improved sharply vs Sol
- ⑤ **Scope control** — OpenAI's own story is “more capable *and* less likely to wander off-policy”

The serious caveat

“Most aligned” arrived with “Critical cyber”

Astra is stronger at finding and chaining exploits. The sharpest cyber features are gated (Daybreak-style programs). That pairing is the policy story of this release, not a footnote.

- Monitoring is harder if the model compresses or conceals intermediate reasoning
- OpenAI added more CoT monitoring *because* traces are a fragile safety tool
- “Welcome to the AGI era” is a vendor framing. OpenAI defines AGI as systems that outperform humans at most economically valuable work. That is a claim to test, not a measurement you already have
- Benchmarks can be saturated, contaminated, or run under vendor conditions

A practical decision rule

Use Astra when

The work is long, tool-heavy, ambiguous, or expensive if wrong: multi-hour research, software migration, computer-use workflows, hard STEM, high-stakes professional artifacts.

Stay on 5.6 Terra / Luna (or other cheap models) when

The task is high-volume, well-specified, and cheap to retry: extraction, routing, templated drafts, classification.

Operating rule

Pick the model for the **workflow**, then set reasoning effort. Do not pay frontier prices for autocomplete.

This was not only an OpenAI story

- **Anthropic Claude** — long context, coding agents, Fable / Opus 5.x as Astra's usual comparison set
- **Google Gemini** — native multimodal, million-token context early, Project Astra assistant prototype
- **xAI Grok** — real-time / X-native line through Grok 4.5
- **Open weights** — Llama, Qwen, DeepSeek R1 and successors collapsed the price of “good enough” reasoning

Five industry trends 2024–2026:

- 1 Open models ate the mid-market
- 2 Reasoning became a product tier
- 3 Agents beat chat as the valuable interface
- 4 Token prices collapsed
- 5 Multimodal became table stakes

Two Astras, one confusion

GPT-6 Astra

OpenAI frontier LLM / agent
September 2026 product
API + ChatGPT + cloud
Benchmarks, computer use, cyber gating

Project Astra

Google DeepMind prototype
Camera / glasses / live talk
Universal assistant research
Not the model in this deck

If a slide, article, or vendor says only “Astra,” ask **which lab** before you compare scores.

What to remember

- 1 The transformer next-token engine is still the engine
- 2 Capability compounded from **data + post-training + inference-time loops**, not from a single architectural miracle
- 3 2022 made assistants; 2024 made reasoners; 2025–26 made workers
- 4 Astra is the first OpenAI model sold as if the loop can finish professional jobs
- 5 Trust requires your own evals, scoped tools, and human review on irreversible actions

Autocomplete → assistant → reasoner → agent → collaborator

Sources and reading posture

- OpenAI, *GPT-6 Astra: A new generation of intelligence*, 3 September 2026
- Launch coverage and benchmark tables from contemporaneous reporting
- Standard accounts of pretraining, RLHF, and test-time compute (o1 / R1 era)

How to cite this deck

Treat vendor benchmarks as **primary claims**, independent leaderboards as **cross-checks**, and production metrics on your tasks as the only numbers that decide spend.

End of briefing

GPT-3 → GPT-6 Astra

Same engine. Different product. Much more agency.